



May 2025

CS2222-1- IoT DEVICE SECURITY
(For EC, CS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is NOT a characteristic of IoT systems?
A) Interconnectivity
B) Scalability
C) Robustness
D) High power consumption
2. Which of the following is an example of IoT connectivity?
A) Wi-Fi
B) Temperature sensor
C) Cloud storage
D) Web application
3. Which of the following best defines the Internet of Things (IoT)?
A) A network of interconnected computers
B) A global network connecting smart devices and sensors
C) A collection of web applications
D) A cloud computing service
4. Which of the following is a common vulnerability in IoT systems?
A) Excessive data storage
B) Lack of wireless capabilities
C) Insecure interfaces
D) Use of strong encryption protocols
5. Which of the following is a symmetric encryption algorithm?
A) RSA
B) DES
C) ECC
D) DSA
6. Which of the following is a fundamental principle of public key cryptography?
A) The same key is used for both encryption and decryption
B) Different keys are used for encryption and decryption
C) It relies on the use of passwords
D) It only works with biometric data
7. What does CBC stand for in the context of block cipher modes of operation?
A) Cipher Block Chaining
B) Code Block Cipher
C) Common Block Cipher
D) Cipher Block Control
8. Which algorithm is primarily used for securing email and other data transmissions through public key cryptography?
A) AES
B) SHA-256
C) DES
D) RSA
9. What is a common risk associated with using default passwords on IoT devices?
A) Increased device speed
B) Higher power consumption
C) Easy exploitation by attackers
D) Improved data integrity
10. Which type of attack uses environmental changes to disrupt the operation of IoT devices?
A) Physical attack
B) Semi invasive attack
C) Logical attack
D) Software based attack
11. Which of the following is considered a noninvasive attack on an IoT device?
A) Power analysis attack
B) Microprobing
C) Fault injection
D) Sidechannel attack

12. Which of the following is a method used to mitigate risks in IoT environments?
 A) Implementing strong access control measures
 B) Disabling encryption
 C) Using default passwords
 D) Ignoring firmware updates
13. What is the main purpose of implementing threat mitigation strategies in IoT?
 A) To eliminate all possible threats
 B) To identify new features for IoT devices
 C) To increase system complexity
 D) To reduce the impact and likelihood of threats
14. Which term describes the process of determining the value of different assets within an IoT environment?
 A) Threat assessment
 B) Asset valuation
 C) Risk analysis
 D) Vulnerability assessment
15. Which of the following is a key component of a secure software development life cycle (SDLC)?
 A) Code obfuscation
 B) Ignoring vulnerability reports
 C) Regular security audits and code reviews
 D) Limiting user access
16. Which of the following is NOT a common threat mitigation measure in IoT security?
 A) Encryption of data
 B) Regular firmware updates
 C) Using strong authentication mechanisms
 D) Disabling firewalls
17. What does TEE stand for in the context of platform security?
 A) Trusted Execution Environment
 B) Technical Execution Engine
 C) Total Encryption Environment
 D) Trusted Encryption Endpoint
18. Which security feature is commonly found in secured MCUs?
 A) Inbuilt web browser
 B) Hardware based cryptographic engine
 C) Wireless charging capability
 D) Larger storage capacity
19. Secure elements come in which of the following form factors?
 A) USB token
 B) Embedded chip
 C) SIM card
 D) All of the above
20. What is one of the key features of Optiga Trust X?
 A) High resolution display
 B) Wireless data transmission
 C) Hardware based key management
 D) Built in camera

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the evolution of IoT from its early roots to its current transformative state, highlighting key technological advancements.	8	L*2	1	1
	b) What is symmetric cryptography and how does it relate to AES CBC and ECB mode?	8	L2	2	1
2.	a) Enumerate the significance of availability in IoT. List the key considerations for ensuring availability in IoT? Discuss the future trends and challenges in ensuring availability in IoT.	8	L2	1	1
	b) What is Elliptic Curve Cryptography? List and explain different components of ECC	8	L2	2	1
3.	a) Discuss the components of Symmetric encryption. Explain monoalphabetic cipher by considering the key "ANOTHER" as an example.	8	L2	1	1
	b) Define Cryptosystem. Explain Caesar cipher. Encrypt the message "work patiently" using Caesar cipher encryption.	8	L2	2	1

Unit – II

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| 4. a) | Explain logical attacks in IoT security and how they exploit weaknesses in software, firmware, or network protocols. Then, list strategies for preventing these attacks in IoT devices. | 8 | L2 | 3 | 1 |
| b) | What is some device-level security concepts that can be used to secure IoT devices? | 8 | L2 | 4 | 1 |
| 5. a) | Identify as how the attack surface be minimized in embedded systems through safeguarding unused hardware components, and what are the key practices involved in proper decommissioning and disabling of such components? | 8 | L2 | 3 | 1 |
| b) | How does a True Random Number Generator (TRNG) differ from pseudo-random number generators (PRNGs), and what are the components and key characteristics of a TRNG architecture? | 8 | L2 | 4 | 1 |
| 6. a) | Applying on an example, define low-level design threat modeling and how can it be used to improve IoT security? | 8 | L2 | 4 | 1 |
| b) | What is the relationship between device-level and hardware-level security concepts in securing IoT devices? | 8 | L2 | 3 | 1 |

Unit – III

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| 7. a) | Define and describe the working of ARM TrustZone in conjunction with Trusted Firmware-M(TF-M) to improve IoT security? | 8 | L2 | 5 | 1 |
| b) | How can hardware-based security solutions like TEE and secured MCUs be integrated with software-based security solutions to enhance IoT security? | 8 | L2 | 5 | 1 |
| 8. a) | What is a secured MCU, Provide a brief explanation regarding its key security capabilities. | 8 | L2 | 5 | 1 |
| b) | What are some of the challenges associated with implementing secured MCUs and secured elements in IoT devices? | 8 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
